AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	CyrusOne SAT10 & SAT11, TX -- station KSSF, America/Chicago
Committed pair	OSM 1533148752 -> 1344369111 OSM way 1533148752 / OSM way 1344369111
Facade gap	246.6 m between the two halls
Weather record	43,628 real hourly records from KSSF
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2263 C at 150 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-08 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 15 of 24 hours with 2 mode changes. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	15 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	15 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	6.209	18.0	7.780	4.597
01	FREE	yes	-	6.749	18.0	6.110	4.050
02	FREE	yes	-	7.296	18.0	5.560	3.667
03	FREE	yes	-	7.402	18.0	4.440	3.079
04	FREE	yes	-	7.846	18.0	3.890	2.941
05	FREE	yes	-	7.859	18.0	2.780	2.410
06	FREE	yes	-	8.402	18.0	2.220	2.539
07	FREE	yes	-	7.443	18.0	2.220	2.345

08	FREE	yes	-	8.982	18.0	6.110	2.334
09	FREE	yes	-	11.758	18.0	8.330	4.220
10	FREE	yes	-	14.062	18.0	8.890	5.131
11	FREE	yes	-	15.130	18.0	10.560	5.608
12	FREE	yes	-	16.311	18.0	12.220	5.867
13	FREE	yes	-	16.760	18.0	13.330	5.739
14	FREE	yes	-	17.863	18.0	13.890	3.928
15	mechanical	no	dry-bulb	18.412	18.0	13.890	3.632
16	mechanical	yes	-	16.860	18.0	12.780	3.288
17	mechanical	yes	-	16.297	18.0	10.560	3.346
18	mechanical	yes	switch budget	15.634	18.0	9.440	3.561
19	mechanical	yes	switch budget	14.052	18.0	7.780	3.454
20	mechanical	yes	switch budget	12.428	18.0	6.670	3.332
21	mechanical	yes	switch budget	11.894	18.0	6.110	4.142
22	mechanical	yes	switch budget	10.718	18.0	5.000	4.822
23	mechanical	yes	switch budget	8.464	18.0	3.890	4.731

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.209 C, which is 11.791 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.597 C, and the margin is measured, not chosen: 4.509 C of group-conditional forecast error for this hour of day, plus 0.0888 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.749 C, which is 11.251 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.050 C, and the margin is measured, not chosen: 3.971 C of group-conditional forecast error for this hour of day, plus 0.0791 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.296 C, which is 10.704 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.667 C, and the margin is measured, not chosen: 3.591 C of group-conditional forecast error for this hour of day, plus 0.0760 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.402 C, which is 10.598 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.079 C, and the margin is measured, not chosen: 2.907 C of group-conditional forecast error for this hour of day, plus 0.1720 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.846 C, which is 10.154 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.941 C, and the margin is measured, not chosen: 2.875 C of group-conditional forecast error for this hour of day, plus 0.0659 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.860 C, which is 10.140 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.410 C, and the margin is measured, not chosen: 2.331 C of group-conditional forecast error for this hour of day, plus 0.0795 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.403 C, which is 9.597 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.539 C, and the margin is measured, not chosen: 2.466 C of group-conditional forecast error for this hour of day, plus 0.0725 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.443 C, which is 10.557 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.345 C, and the margin is measured, not chosen: 2.137 C of group-conditional forecast error for this hour of day, plus 0.2075 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.982 C, which is 9.018 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.334 C, and the margin is measured, not chosen: 2.252 C of group-conditional forecast error for this hour of day, plus 0.0818 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.758 C, which is 6.242 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.220 C, and the margin is measured, not chosen: 4.122 C of group-conditional forecast error for this hour of day, plus 0.0979 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.062 C, which is 3.938 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.131 C, and the margin is measured, not chosen: 4.959 C of group-conditional forecast error for this hour of day, plus 0.1720 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.130 C, which is 2.870 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.608 C, and the margin is measured, not chosen: 5.478 C of group-conditional forecast error for this hour of day, plus 0.1296 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.311 C, which is 1.689 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.867 C, and the margin is measured, not chosen: 5.667 C of group-conditional forecast error for this hour of day, plus 0.2009 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.760 C, which is 1.240 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.739 C, and the margin is measured, not chosen: 5.638 C of group-conditional forecast error for this hour of day, plus 0.1004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.863 C, which is 0.137 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.928 C, and the margin is measured, not chosen: 3.845 C of group-conditional forecast error for this hour of day, plus 0.0826 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.412 C against a 18.0 C limit, so it fails by 0.412 C. It would take a limit 0.412 C higher, or a bound 0.412 C tighter, to change this hour.

16:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.860 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.296 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.634 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.052 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.429 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.894 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.718 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.464 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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